

Joseph Steven (Steve) Ferguson

joelemmadad@gmail.com • (360)731-2366 • Bremerton, WA

Summary

Innovative Senior **Software** Engineer with over 20 years of experience developing reliable solutions to complex problems. Proficient with a variety of programming languages, **technologies** and frameworks, and adept at utilizing **new technologies**. Experienced in collaborating across teams and mentoring developers to enhance **code** efficiency, reliability, and maintainability.

Technical Skills

- **Programming Languages:** **Java**, Kotlin, SQL, JavaScript, C#
- **Web Technologies:** HTML, CSS, React.js, **Microservices**, REST, GraphQL, SOAP
- **Databases:** AWS DynamoDB, MongoDB, Postgres, Mysql, SQL Server, Oracle
- **Streaming/Messaging:** **Kafka**, SQS, Rabbit MQ, JMS, AWS Firehose, AWS Lambda
- **Tools & Platforms:** Unix, .NET, **Spring**, Flink/Statefun, AWS, **Kubernetes**, **Docker**, Git, Datadog, Jenkins, Spinnaker
- **Methodologies:** Object Oriented, **Agile**, **Scrum**, CI/CD

Experience

Experia Group, Seattle, WA • Senior **Software** Engineer

2015 - 2024

- Led a team of 4 engineers to design and deliver TravelAds on Expedia Partner Central, contributing to a significant increase in advertising revenue.
 - Developed a microservice in Kotlin with **Spring** Boot that consumes campaign **data** from **Kafka**, persists it in AWS DynamoDB, and exposes advertising status via a GraphQL API.
 - Built and managed a Spinnaker CI/CD pipeline for deploying to AWS EKS, and created a Datadog dashboard for monitoring performance and ensuring system reliability.
 - Collaborated cross-functionally with product managers, platform architects, UX designers, and engineers to deliver the **project** in under 3 months.
 - Led design and **code** reviews to maintain high-**quality engineering** standards.
- Spearheaded the **development** of a financial platform monitoring system, supporting over 1,000 events per second across distributed **services** with a team of 7 engineers.
 - Architected a scalable **microservices** system in **Java/Spring** Boot with AWS SQS and DynamoDB to track financial transaction workflows from booking to general ledger.

- Built a real-time dashboard with SSO and RBAC in React.js, CSS, and HTML, improving visibility and alerting on platform issues that impacted \$33B in transactions (2019).
 - Collaborated with technical leads, product owners, and stakeholders to ensure seamless integration with a suite of 10+ systems.
 - Led cost optimization efforts, reducing **cloud** resource usage and operational expenses.
 - Mentored engineers and drove adoption of **best practices** through regular **code** reviews and technical guidance.
- Led a team of **8** engineers on core **development** efforts for Expedia and VRBO's tax systems, ensuring accurate and compliant taxation across global markets.
 - Built **services** in **Java**, **Spring**, and Sql Server to support dynamic tax rule application and calculations.
 - Designed a secure, self-service rule **management** interface with SSO and RBAC, enabling tax analysts to author, test, and deploy rules independently—reducing dependency on **engineering** and support.
 - Developed an API integration allowing VRBO to leverage Expedia's tax system, resolving compliance issues and eliminating fines in France.
 - Collaborated closely with analysts, **project** managers, and stakeholders to ensure systems aligned with evolving business and regulatory requirements.
- Prototyped a real-time stream processing system using **Apache** Flink with StateFun on AWS EMR, enabling complex joins over **Kafka** streams.
 - Developed a persistence mechanism that supported long term state persistence to MongoDB.
 - The **architecture** was adopted across the organization and became a blueprint for future real-time **data** solutions.

JP Morgan Chase, Seattle, WA • Applications **Developer **Lead****

2008 - 2015

- Developed, maintained and supported .NET libraries used on chase.com in C# and JavaScript.
- Collaborated with technical leads and operations staff to diagnose and repair customer facing issues on chase.com.
- Developed the Login module currently used on chase.com in JavaScript.
- Reviewed **code** written and consulted developers on security concerns as Security Champion for the Seattle office.

Washington Mutual, Seattle, WA • **Software Architect**

2002 - 2008

- Developed system **architecture** documentation and reviewed documentation developed by others. Collaborating with **project** managers, other architects, and **lead** developers to ensure the **architecture** met the **project** objectives, used **best practices**, and was supported by the **development** staff.

- Developed C# .Net **web** components allowing wamu.com customers to pay bills, transfer funds, and send/receive private messages.
- Enhanced **Java** based real time trading application interfacing with Tradeweb and Bloomberg trading systems
- Developed price publishing system to allow market **data** from multiple sources to be collected, fed into different pricing systems and published to multiple destinations in real-time. The system utilized JMS (WebLogic), **Spring**, and Reuters APIs.

Keystroke Technology Solutions, Seattle WA • Principal Architect

1997 - 2002

- **Lead** the **development** of **internet**-based mortgage systems using **J2EE architecture**.
- Designed and implemented **java** based security systems, credit reporting systems, and a rule **management** and decisioning system for determining mortgage eligibility and pricing.
- Helped recruit developers, and mentored less experienced developers.

Olympic Peninsula Consulting, WA • Computer System Consultant

1993 - 1997

- Developed and maintained financial, government, and hospitality systems as an independent contractor.
- Created and maintained **internet** monitoring **software**, reporting systems, and call accounting **software** for various clients.

Computer Science Corporation, Keyport WA • Member of Advisory Staff

1989 - 1993

- Developed **software** for network security, X.500 Directories, and SMTP mail gateways.
- Ported and maintained TCP/IP network **software** and developed GUI components for Unix environments.

Education

- **Western Washington University, Bellingham WA** *1984 - 1986*
B.S. Computer Science
- **Olympic College, Bremerton WA** *1982 - 1984*
A.S. Computer Science

Joseph Ferguson

- Bremerton, WA, US

Contact Information

- ek5-m95-9uf@mail.dice.com
- 3607312366

Work History

- Senior **Software** Engineer Expedia Group

Skills

- **structured query language** - 24 years
- **database** - 23 years
- **html** - 23 years
- **java** - 23 years
- **cascading style sheets** - 22 years
- **object-oriented programming** - 21 years
- **javascript** - 16 years
- **spring framework** - 11 years
- **microservices** - 10 years
- **react** - 10 years
- **spring boot** - 10 years
- **amazon web services** - 9 years
- **mysql** - 9 years
- **representational state transfer** - 8 years
- **git** - 6 years
- **kotlin** - 1 years
- **software** - 24 years
- **unix** - 22 years
- **leadership** - 21 years
- **apache http server** - 20 years
- **reporting** - 18 years
- **software development** - 16 years

- **api** - 15 years
- **spring** - 15 years
- **mentorship** - 14 years
- **code review** - 10 years
- **mentoring** - 10 years
- **role-based access control** - 10 years
- **agile** - 9 years
- **amazon dynamodb** - 9 years
- **amazon sqs** - 9 years
- **apache kafka** - 9 years
- **jenkins** - 9 years
- **scrum** - 9 years
- **ui** - 9 years
- **ux** - 9 years
- **docker** - 8 years
- **continuous integration and continuous** - 6 years
- **microservices architecture** - 6 years
- **kubernetes** - 3 years
- **performance monitoring** - 3 years
- **soap** - 3 years
- **apache flink** - 2 years
- **amazon elastic kubernetes service** - 1 years
- **datadog** - 1 years
- **graphql** - 1 years
- **postgresql** - 1 years
- **spinnaker** - 1 years
- **cost optimization** - 2 years
- **mongodb** - 2 years
- **system architecture** - 17 years
- **amazon web services simple queue** - 3 years
- **single sign-on** - 3 years
- **amazon web services lambda** - 1 years
- **microsoft sql server** - 10 years
- **rabbitmq** - 3 years
- **c#** - 15 years
- **.net** - 13 years
- **.net framework** - 8 years
- **java platform, enterprise edition** - 6 years
- **pricing** - 11 years
- **java message service** - 7 years
- **oracle weblogic server** - 7 years
- **oracle database** - 4 years
- **c** - 8 years

Work Preferences

- Desired Work Settings: Remote or Hybrid
- Likely to Switch: True
- Willing to Relocate: False
- Travel Preference: 0%
- Work Authorization:
 - US
- Work Documents:
 - US Citizen
- Security Clearance: False
- Third Party: False
- Employment Type:
 - Full-time
 - Part-time
 - Contract - Independent
 - Contract - W2

Profile Sources

- linkedin: <https://www.linkedin.com/in/steve-ferguson-a7a141189>
- Dice: <https://www.dice.com/employer/talent/profile/6d70beeff350b463fabd839c3c2764b0>